

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

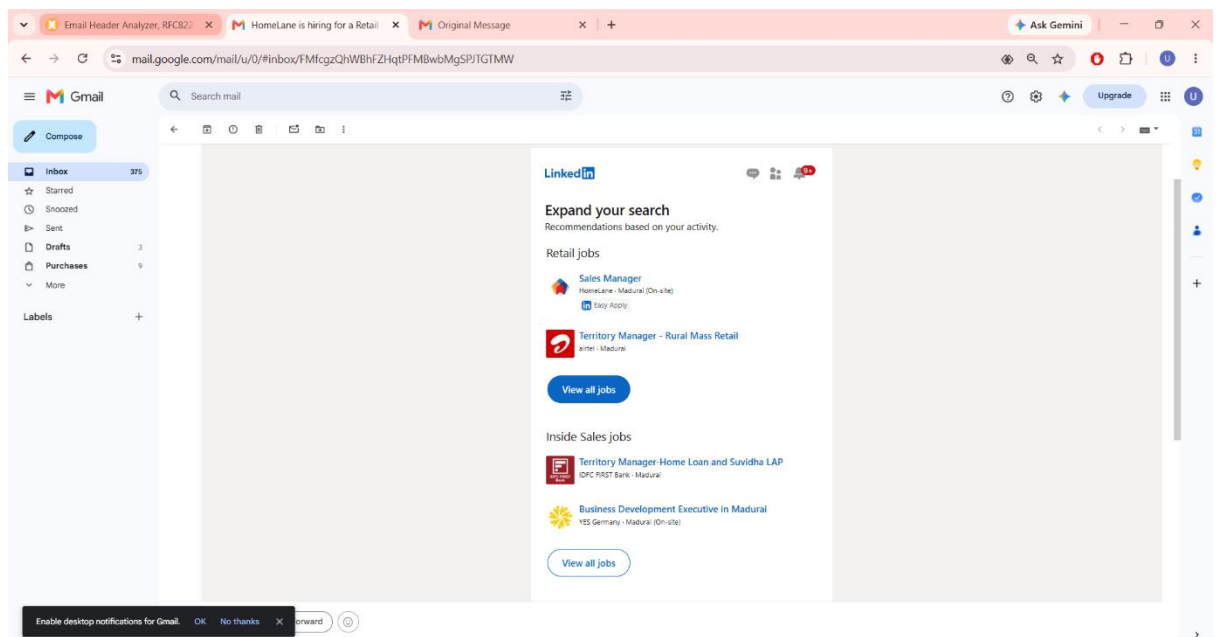
Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.





Original Message

Message ID	<1508452193.8453086.1787214488390@ltx1-app42059.prod.linkedin.com>
Created at:	Thu, Aug 20, 2026 at 1:58 PM (Delivered after 0 seconds)
From:	LinkedIn <jobs-noreply@linkedin.com>
To:	Uday Kumar Reddy Mooli <mooliudaykumarreddy77@gmail.com>
Subject:	airtel is hiring for a Telecom role
SPF:	PASS with IP 108.174.0.155 Learn more
DKIM:	'PASS' with domain linkedin.com Learn more
DMARC:	'PASS' Learn more

[Download Original](#)

mxtoolbox.com/Public/Tools/EmailHeaders.aspx?huid=d637634a-47cc-4425-957e-a536c23dd465

Header Analyzed
Email Subject: HomeLane is hiring for a Retail role

Delivery Information

- DMARC Compliant
- SPF Alignment
- SPF Authenticated
- DKIM Alignment
- DKIM Authenticated

Relay Information

Received Delay: 1 seconds

From mail-ca.linkedin.com to mx.google.com

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	-	mail-ca.linkedin.com 108.174.6.149	mx.google.com	ESMTPS	8/24/2026 8:28:09 AM	✓
2	1 Second		2002:a17:504:29c7:b0:1e80:4800:885a	SMTP	8/24/2026 8:28:10 AM	

Email Header Analyzer, RFC822 x HomeLane is hiring for a Retail x Original Message x + Ask Gemini

mxttoolbox.com/Public/Tools/EmailHeaders.aspx?huid=d637634a-f7cc-4425-957e-a536c23dd465

EmailHeaders advertisement image

SPF and DKIM Information

dmARC:linkedin.com Show Solve Email Delivery Problems

v=DMARC1; p=reject; rua=mailto:d@rue.agari.com,mailto:ryf3q-9358@rue.dmarc.emailanalyst.com; ruf=mailto:d@rue.agari.com

SPF:bounce.linkedin.com:108.174.6.149 Show Solve Email Delivery Problems

vs=spf1 redirect=_spf.linkedin.com

DKIM:mailb.linkedin.com:d2048-202308-0b Show

DKIM Public Record

v=DKIM1; k=russa; p=H1E1Sj40BgkqH10hWBAQEFAAOCQBAWIEIBCgKCAQEazIcVQfTSoxH07SRQ/Rq@gh6ShwKpKvQq111105glenRufHqknyHfUzIzonRn/Na2012Nz2Jf0rVhrcz57Vp0Cg7m0PFD/0t4csFdw051m0Hv18AyL5izeGIX0Ac2Nf0so3u38q3ESKldRk0WpQZJSDKagD8Uk1zKyf6ef3V513kE4Hw

DKIM Signature

v=1; a=russa-sha256; c=relaxed/relaxed; d=mailb.linkedin.com; s=d2048-202308-0b; h=1list-unsubscribe-post:1list-unsubscribe:x-linkedIn-fblix-linkedIn-template: x-linkedIn-class:to:content-type:mime-version:subject:message-id:from:date: from; bh=0/0b

DKIM:linkedin.com:d2048-202308-00 Show

DKIM Public Record

v=DKIM1; k=russa; p=H1E1Sj40BgkqH10hWBAQEFAAOCQBAWIEIBCgKCAQEazIcVQfTSoxH07SRQ/Rq@gh6ShwKpKvQq111105glenRufHqknyHfUzIzonRn/Na2012Nz2Jf0rVhrcz57Vp0Cg7m0PFD/0t4csFdw051m0Hv18AyL5izeGIX0Ac2Nf0so3u38q3ESKldRk0WpQZJSDKagD8Uk1zKyf6ef3V513kE4Hw

Exp_4 99240040292.pdf x WHOIS - Search x Whois IP 108.174.6.149 +

https://www.whois.com/whois/108.174.6.149

Gmail YouTube Maps Technology Platfor... C Tutorial - Tutoria... Sponsored Ad - Su... Mafia 1 - Free Dow... 3:37 Now playing...

Whois Identity for everyone Domains Hosting Websites Email Security Whois Deals

108.174.6.149

Whois IP 108.174.6.149

Updated 1 second ago

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#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange: 108.174.0.0 - 108.174.15.255
CIDR: 108.174.0.0/20
NetName: LINKEDIN
NetHandle: NET-108-174-0-0-1
Parent: NET108 (NET-108-0-0-0-0)
NetType: Direct Allocation
OriginAS:
Organization: LinkedIn Corporation (LINKE-1)
RegDate: 2012-01-11
Updated: 2021-12-14
Ref: https://rdap.arin.net/registry/ip/108.174.0.0

OrgName: LinkedIn Corporation
OrgId: LINKE-1
Address: 1000 W Maude Ave
```